



Signal Transduction

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Learning outcome

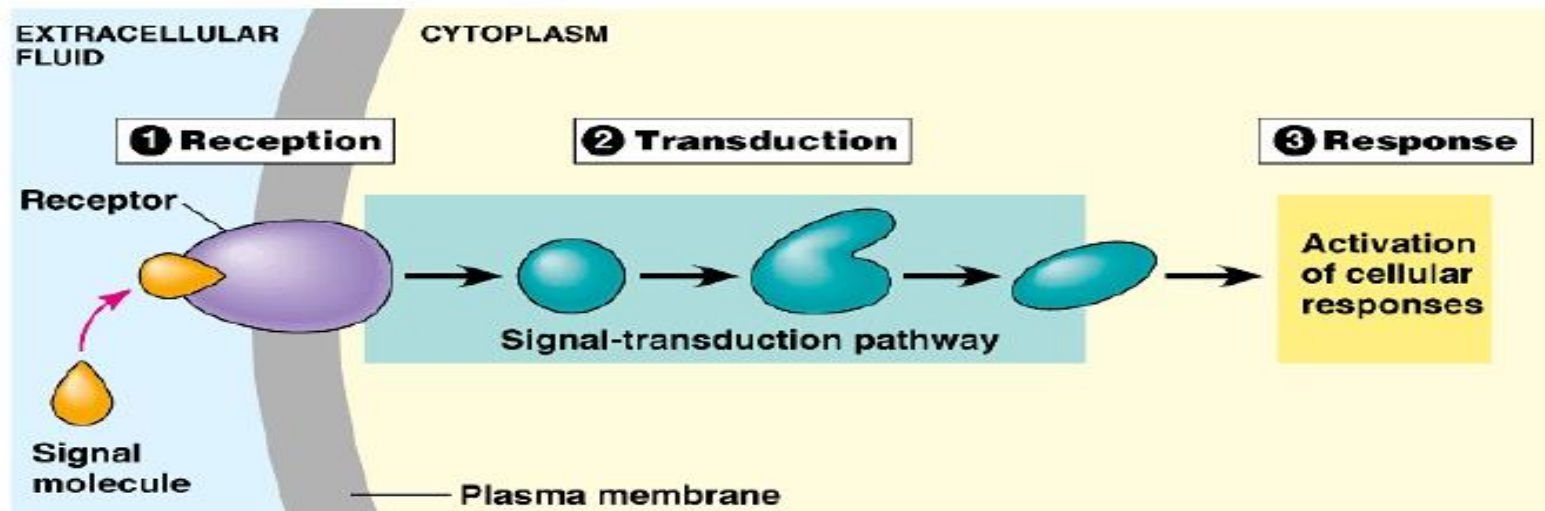
By the end of this lecture, audiences will be able to:

- Define the Signal Transduction.
- Identify the forms of cellular interactions.
- Describe the steps of cellular signaling.

Introduction

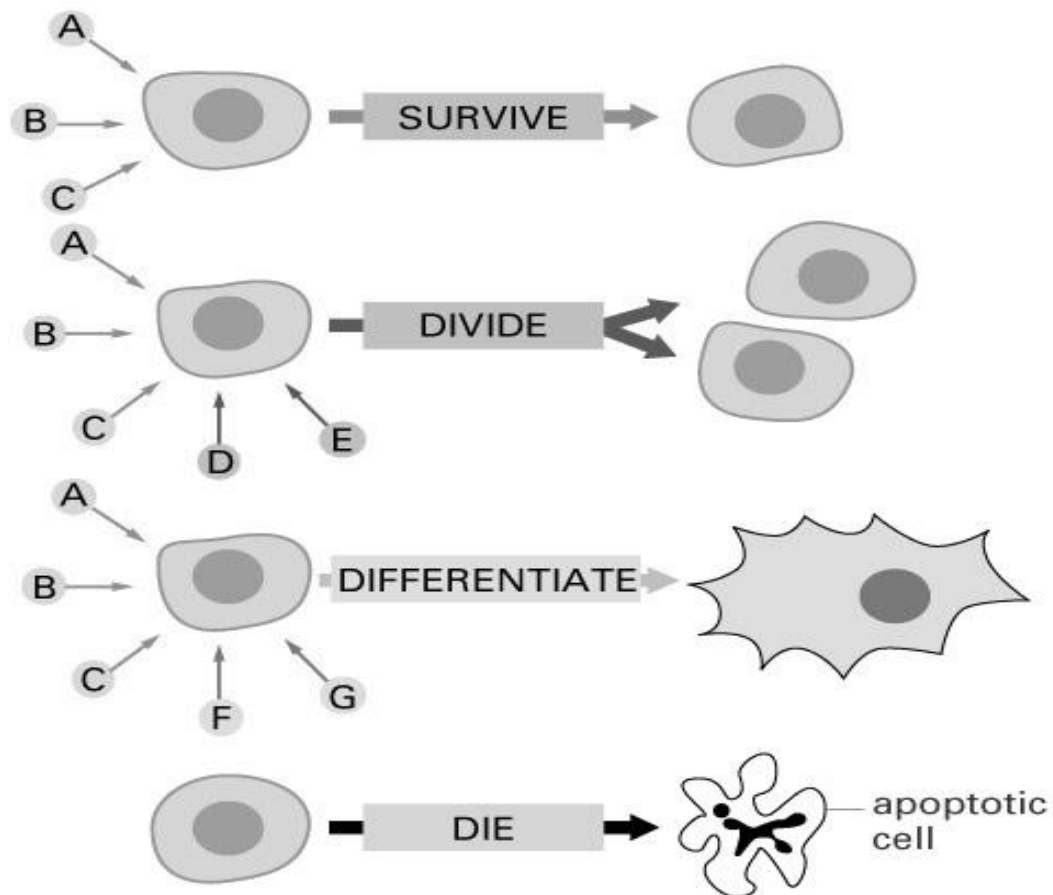
- It's also termed cell-cell communication or cell-cell interaction
Or cell-cell messaging or cell-cell talking.
- The first step is the binding of the signaling molecules and the receptors.
- This binding causes receptor activation (conformational changes).
- Signal transduction, the “signal” is the receptor activation, and “transduction” denotes the process by which a stimulus is transformed into a response.

Three Stages of Signal Transduction



Forms of cellular response

- (1) The permeability, transport properties, or electrical state of the plasma membrane.
- (2) Metabolism
- (3) Secretory activity
- (4) Rate of proliferation and differentiation
- (5) Contractile or other activities.



Importance of cellular signaling:

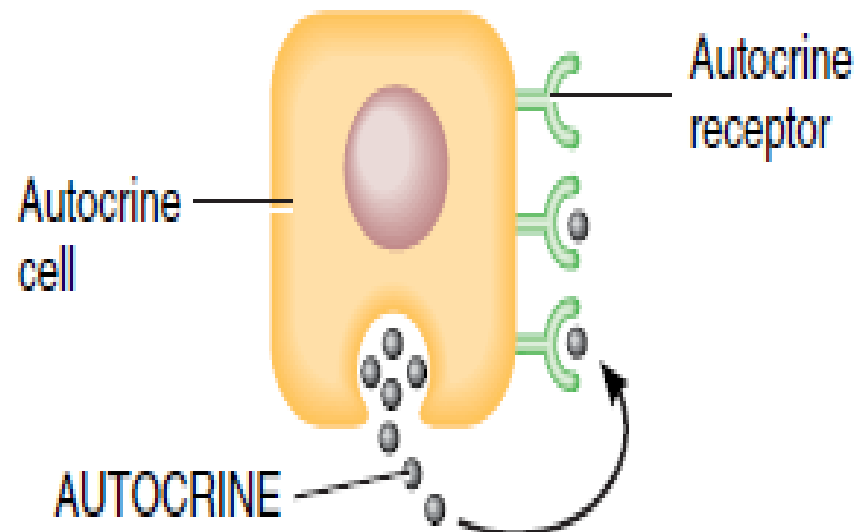
- For a coordinated function of cells in a tissue, tissues in an organ, organs in a system and systems in the body, cells need to be able to communicate with each other.
- Each cell should be capable of sending chemical signals to other cells and of receiving chemical signals from other cells.
- To allow our body functions as an integrated whole, rather than as a massive collection of individual cells acting independently.

Types of signaling

- A cell can communicate signals to other cells in various ways.
- **Autocrine signaling**
- **Paracrine signaling**
- **Endocrine signaling**
- **Direct signaling**
- **Synaptic signaling**

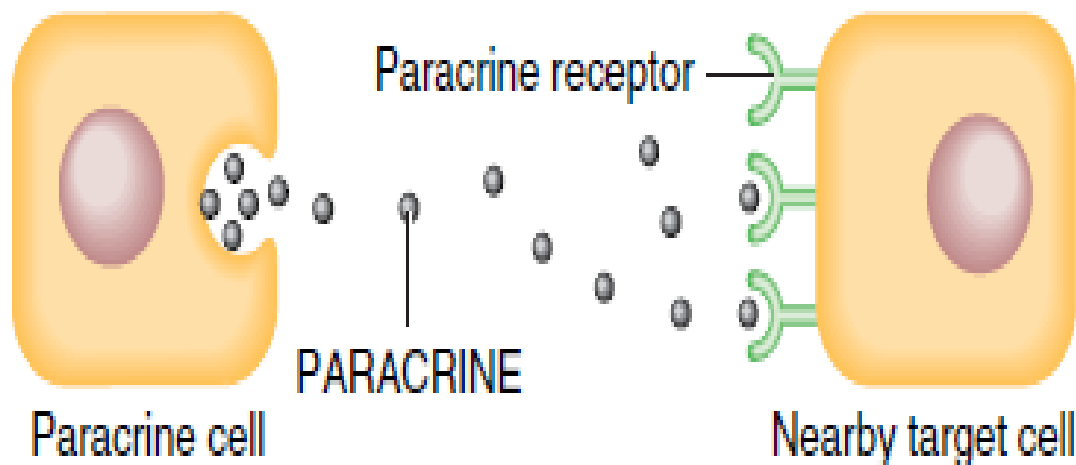
Autocrine signaling

- The signaling cells release the messengers (signaling molecules) that act on the same cell that secreted them.
- Interleukin-2 (IL-2), which is released by helper T cells (a type of white blood cell) during immune responses and acts on the same cell that released it to proliferate.



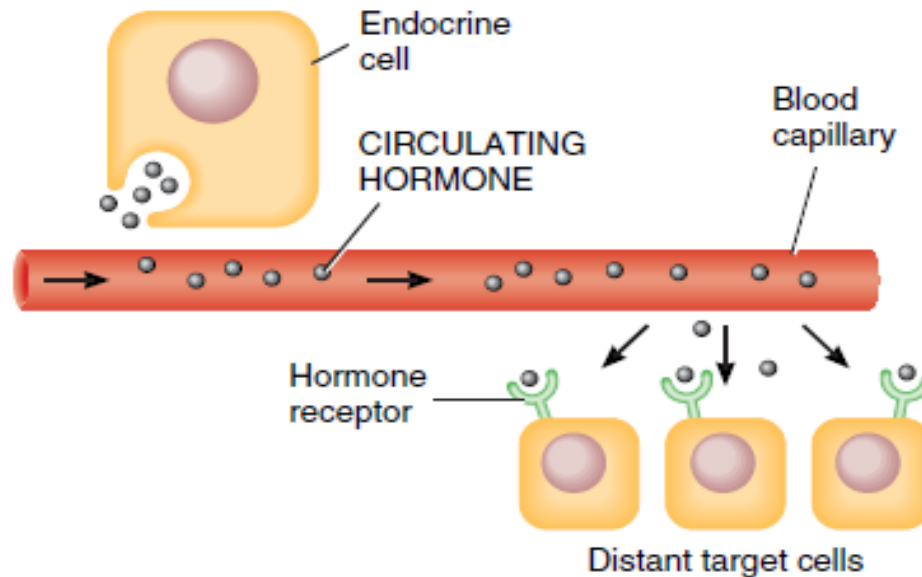
Paracrine signaling

- Is a way for a cell to affect the behavior of **neighboring cells** by secreting chemicals into the common intercellular space.
- Gas nitric oxide (NO), which is released by endothelial cells lining blood vessels and causes relaxation of nearby smooth muscle.



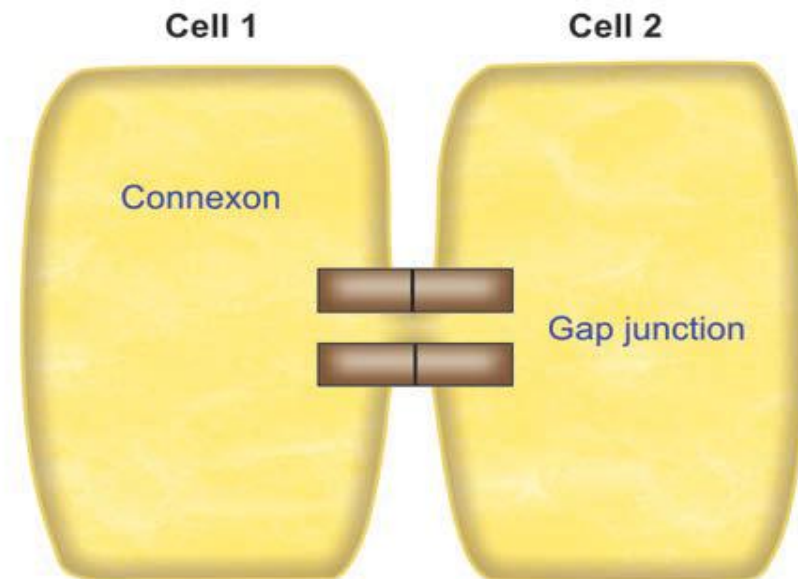
Endocrine signaling

- A cell secretes chemicals into the bloodstream. Those chemicals affect the behavior of **distant target cells**.
- Thyroid hormones, growth hormone....etc



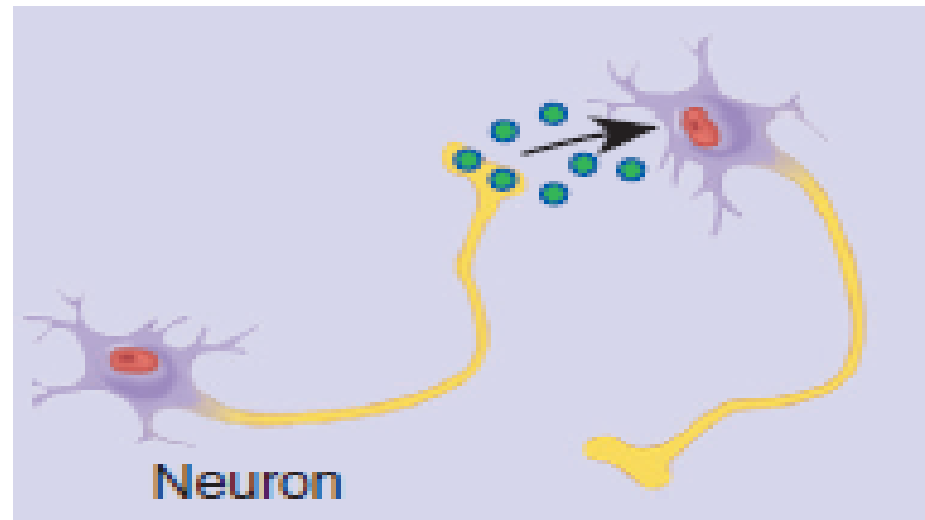
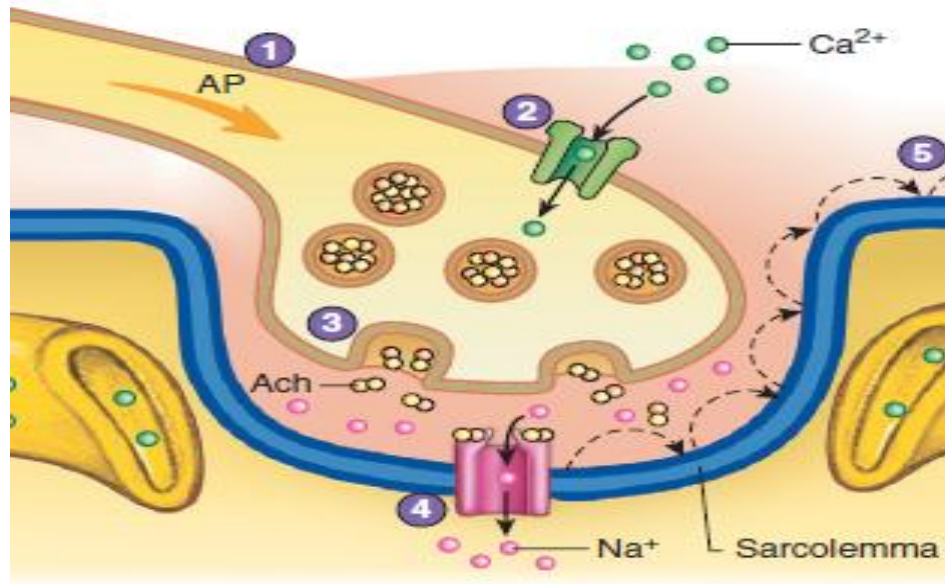
Direct signaling

- Is a transfer of ions or small molecules from one cell to its neighbor through pores in the membrane.
- Those pores are built out of membrane proteins and are called **gap junctions**.
- This is the fastest mode of cell-cell communication.
- An example of this process can be found in the heart.



Synaptic signaling

- Is found in the nervous system.
- It is a highly specific and localized type of paracrine signaling between two nerve cells or between a nerve cell and a muscle cell.
- E.g: Acetylcholine



How does a cell receive a signal?

- Some molecules are hydrophilic (water soluble) they are incapable of entering the cell through the plasma membrane.
- Some small hormones enter the cell directly, by passing through the membrane.
- Examples are steroid hormones and thyroid hormones.....

